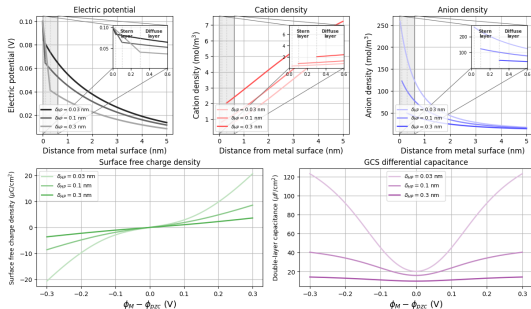




EXERCISE 3

Electric Double Layer Modeling: Gouy–Chapman–Stern & Bikerman

May 21, 2026 | O. Schmidt | Theory and Computation of Energy Materials (IET-3)

AIM OF TODAY'S EXERCISES

Today, we investigate the Gouy–Chapman–Stern and Bikerman model.
Some help can be found on

→ Exercises ←

Objectives: To understand the strengths and limitations of the Gouy–Chapman–Stern and Bikerman model with respect to

- Electric potential
- Anion and Cation concentration profile
- Surface Free Charge Density
- Double Layer Capacitance

Furthermore, we provide another model corresponding to changing the Bikerman model, using different size coefficients.

GOUY–CHAPMAN–STERN: FUNDAMENTAL CONSTANTS

Symbol	Value	Unit	Meaning
N_A	6.022×10^{23}	mol^{-1}	Avogadro constant
k_B	8.617×10^{-5}	eV K^{-1}	Boltzmann constant
e_0	1.602×10^{-19}	C	Elementary charge
ε_0	8.854×10^{-12}	F m^{-1}	Vacuum permittivity

GOUY–CHAPMAN–STERN: MODEL PARAMETERS

Symbol	Unit	Meaning
$\phi_M - \phi_{pzc}$	V	Applied potential relative to the Potential of Zero Charge.
T	K	Temperature
ϵ_S	–	Relative permittivity of the solution region
c_b	mM	Bulk ion concentration
ϵ_{HP}	–	Relative permittivity of the compact Helmholtz layer
δ_{HP}	nm	Distance from metal surface to Helmholtz plane

MODEL SETUP: GOUY–CHAPMAN–STERN

The Gouy–Chapman–Stern model combines a compact Helmholtz layer with a diffuse Gouy–Chapman layer.

The compact Helmholtz layer is described by a linear potential drop, while the diffuse layer is described by the Gouy–Chapman solution starting from the Helmholtz-plane potential ϕ_{HP} .

Thus, the total potential is constructed from two parts:

$$\phi^{(\text{GCS})}(x) = \begin{cases} \phi_{\text{H}}(x), & 0 \leq x \leq \delta_{\text{HP}} \\ \phi_{\text{GC}}(x), & x > \delta_{\text{HP}} \end{cases}$$

The value ϕ_{HP} is not chosen freely. It is determined by coupling the Helmholtz layer and the Gouy–Chapman diffuse layer through their surface charge densities.

MODEL SETUP: HELMHOLTZ LAYER

In the compact Helmholtz layer, there are no mobile ions. Therefore, the potential satisfies Laplace's equation.

Helmholtz layer: $0 \leq x \leq \delta_{\text{HP}}$

$$\frac{d^2\phi_{\text{H}}}{dx^2} = 0$$

$$\phi_{\text{H}}(0) = \phi_{\text{M}} - \phi_{\text{pzc}}$$

$$\phi_{\text{H}}(\delta_{\text{HP}}) = \phi_{\text{HP}}$$

The Helmholtz-plane potential ϕ_{HP} is still unknown at this point.

MODEL SETUP: POTENTIAL DROP SPLIT

The total potential drop is split into a compact Helmholtz contribution and a diffuse Gouy–Chapman contribution:

Potential drop

$$\phi_M - \phi_{pzc} = \underbrace{[(\phi_M - \phi_{pzc}) - \phi_{HP}]}_{\text{Helmholtz drop}} + \underbrace{\phi_{HP}}_{\text{Gouy–Chapman drop}}$$

The Helmholtz model alone does not determine ϕ_{HP} . The value of ϕ_{HP} is fixed by matching the Helmholtz and Gouy–Chapman surface charge densities.

MODEL SETUP: CHARGE-DENSITY COUPLING

The Helmholtz-plane potential ϕ_{HP} is determined from

$$\sigma_{\text{H}} ((\phi_{\text{M}} - \phi_{\text{pzc}}) - \phi_{\text{HP}}) = \sigma_{\text{GC}} (\phi_{\text{HP}} - 0)$$

Helmholtz charge density

$$\sigma_{\text{H}} ((\phi_{\text{M}} - \phi_{\text{pzc}}) - \phi_{\text{HP}}) = \varepsilon_0 \varepsilon_{\text{HP}} \frac{(\phi_{\text{M}} - \phi_{\text{pzc}}) - \phi_{\text{HP}}}{\delta_{\text{HP}}}$$

Gouy–Chapman diffuse-layer charge density

$$\sigma_{\text{GC}} (\phi_{\text{HP}}) = \sqrt{8k_{\text{B}} T \varepsilon_0 \varepsilon_{\text{S}} N_{\text{A}} c_{\text{b}}} \sinh \left(\frac{e_0 \phi_{\text{HP}}}{2k_{\text{B}} T} \right)$$

MODEL SETUP: GOUY–CHAPMAN CONCENTRATIONS

The ion concentrations are only defined in the diffuse solution region.

Gouy–Chapman concentrations for $x > \delta_{\text{HP}}$

$$c_{-}(x) = c_{\text{b}} \exp\left(\frac{e_0 \phi_{\text{GC}}(x)}{k_{\text{B}} T}\right)$$
$$c_{+}(x) = c_{\text{b}} \exp\left(-\frac{e_0 \phi_{\text{GC}}(x)}{k_{\text{B}} T}\right)$$

The compact Helmholtz layer contains no mobile cations or anions in this model.

MODEL SETUP: GOUY–CHAPMAN–STERN

The Gouy–Chapman–Stern potential is obtained by joining the Helmholtz solution and the Gouy–Chapman solution after ϕ_{HP} has been determined from the surface-charge matching condition.

Gouy–Chapman–Stern: Piecewise structure

$$\phi^{(\text{GCS})}(x) = \begin{cases} \phi_{\text{H}}(x), & 0 \leq x \leq \delta_{\text{HP}} \\ \phi_{\text{GC}}(x), & x > \delta_{\text{HP}} \end{cases}$$

The two functions ϕ_{H} and ϕ_{GC} are written explicitly on the next slide.

MODEL SETUP: EXPLICIT GCS POTENTIAL

Helmholtz and Gouy–Chapman contributions

$$\phi_H(x) = (\phi_M - \phi_{pzc}) - \frac{x}{\delta_{HP}} [(\phi_M - \phi_{pzc}) - \phi_{HP}]$$

$$\phi_{GC}(x) = \frac{4k_B T}{e_0} \operatorname{arctanh} \left[\tanh \left(\frac{e_0 \phi_{HP}}{4k_B T} \right) \exp \left(-\frac{x - \delta_{HP}}{\lambda_D} \right) \right]$$

$$\lambda_D^{-1} = \left(\frac{2e_0^2 N_A c_b}{\varepsilon_0 \varepsilon_S k_B T} \right)^{1/2}$$

SURFACE FREE CHARGE DENSITY: GOUY–CHAPMAN–STERN

The Helmholtz and Gouy–Chapman parts are combined through equality of their surface charge densities.

Charge-density matching condition

$$\sigma_{\text{H}} = \sigma_{\text{GC}}$$
$$-\varepsilon_0 \varepsilon_{\text{HP}} \frac{\phi_{\text{HP}} - (\phi_{\text{M}} - \phi_{\text{pzc}})}{\delta_{\text{HP}}} = -\varepsilon_0 \varepsilon_{\text{S}} \left. \frac{d\phi_{\text{GC}}}{dx} \right|_{x=\delta_{\text{HP}}^+}$$

This equation determines the unknown Helmholtz-plane potential ϕ_{HP} .

DIFFUSE-LAYER CHARGE RELATION

The diffuse-layer contribution can be written in terms of the Helmholtz-plane potential.

Gouy–Chapman diffuse-layer charge density

$$\begin{aligned}\sigma_{\text{GC}} &= -\varepsilon_0 \varepsilon_{\text{S}} \left. \frac{d\phi_{\text{GC}}}{dx} \right|_{x=\delta_{\text{HP}}^+} \\ &= \sqrt{8k_{\text{B}} T \varepsilon_0 \varepsilon_{\text{S}} N_{\text{A}} c_{\text{b}}} \sinh \left(\frac{e_0 \phi_{\text{HP}}}{2k_{\text{B}} T} \right).\end{aligned}$$

Together with $\sigma_{\text{H}} = \sigma_{\text{GC}}$, this fixes ϕ_{HP} for a given applied potential $\phi_{\text{M}} - \phi_{\text{pzc}}$

CAPACITANCE: GOUY–CHAPMAN–STERN

Gouy–Chapman–Stern: A series capacitance analogy

$$\frac{1}{C_{\text{dl}}^{(\text{GCS})}} = \frac{1}{C_{\text{HP}}} + \frac{1}{C_{\text{D}}(\phi_{\text{HP}})}$$

$$C_{\text{HP}} = \frac{\varepsilon_0 \varepsilon_{\text{HP}}}{\delta_{\text{HP}}}$$

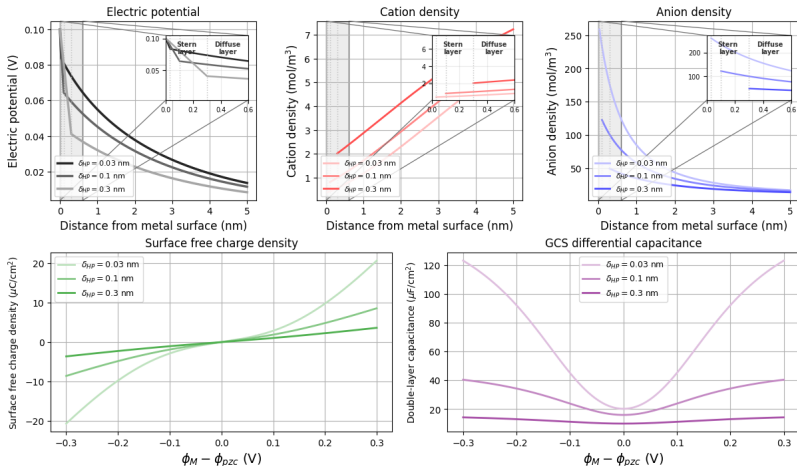
$$C_{\text{D}}(\phi_{\text{HP}}) = \frac{d\sigma_{\text{GC}}(\phi_{\text{HP}})}{d\phi_{\text{HP}}} = \varepsilon_0 \varepsilon_{\text{S}} \lambda_{\text{D}}^{-1} \cosh\left(\frac{e_0 \phi_{\text{HP}}}{2k_{\text{B}} T}\right)$$

QUESTION 1: GOUY–CHAPMAN–STERN

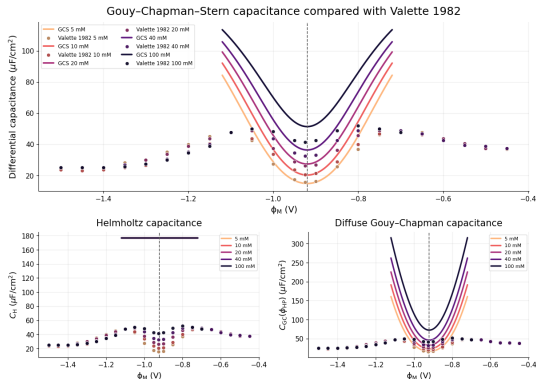
- 1.1) How does the electric potential change when varying the Stern thickness δ_{HP} ?
- 1.2) How does increasing δ_{HP} change the potential drop inside the compact layer?
- 1.3) How do the cation and anion concentrations change when the Stern layer is introduced?
- 1.4) Why must the compact-layer and diffuse-layer surface charge densities be equal?
- 1.5) How is ϕ_{HP} determined from this charge-continuity condition?
- 1.6) Why does the GCS capacitance behave like capacitors in series? What would the capacitance be if we added another layer?

The next two slides provide visual guides for the solution and comparison with experiment.

QUESTION 1: VISUAL SOLUTION GUIDE



QUESTION 1: EXPERIMENTAL COMPARISON



Question: How well does the Gouy–Chapman–Stern capacitance reproduce the experimental Valette data, and where does it start to deviate?

BIKERMAN: FUNDAMENTAL CONSTANTS

Symbol	Value	Unit	Meaning
N_A	6.022×10^{23}	mol^{-1}	Avogadro constant
k_B	8.617×10^{-5}	eV K^{-1}	Boltzmann constant
e_0	1.602×10^{-19}	C	Elementary charge
ε_0	8.854×10^{-12}	F m^{-1}	Vacuum permittivity

BIKERMAN: MODEL PARAMETERS

Symbol	Unit	Meaning
$\phi_M - \phi_{pzc}$	V	Applied potential relative to the Potential of Zero Charge.
T	K	Temperature
ϵ_S	–	Relative permittivity of the solution region
c_b	mM	Bulk ion concentration
r_0	nm	Effective ion / lattice size parameter

MODEL SETUP: BIKERMAN

The Bikerman model modifies the Gouy–Chapman model by including finite ion-size effects.

In the Gouy–Chapman model, ions are point-like. Therefore, the ion concentration can become unrealistically large close to a highly charged surface.

In the Bikerman model, the finite size of the ions is included through a steric denominator. This limits the near-surface ion density.

MODEL SETUP: BIKERMAN VARIABLES

Define the dimensionless potential and finite-size parameter:

Dimensionless variables

$$U(x) = \frac{e_0 \phi(x)}{k_B T}, \quad \nu = 2r_0^3 N_A c_b$$

$$D_B(U) = 1 + 2\nu \sinh^2 \left(\frac{U}{2} \right)$$

The denominator $D_B(U)$ is the finite-size correction. For $r_0 \rightarrow 0$, one obtains $D_B(U) \rightarrow 1$ and the Gouy–Chapman model is recovered.

MODEL SETUP: BIKERMAN EQUATION

The Bikerman model is described by a modified Poisson–Boltzmann equation.

Bikerman: Governing equation for ϕ

$$\varepsilon_0 \varepsilon_S \frac{d^2 \phi}{dx^2} = \frac{2e_0 N_A c_b \sinh\left(\frac{e_0 \phi}{k_B T}\right)}{1 + 4r_0^3 N_A c_b \sinh^2\left(\frac{e_0 \phi}{2k_B T}\right)}$$

with boundary conditions

$$\phi(0) = \phi_M - \phi_{\text{pzc}}, \quad \lim_{x \rightarrow \infty} \phi(x) = 0$$

MODEL SETUP: BIKERMAN CONCENTRATIONS

The finite-size correction modifies the Boltzmann concentrations by the same denominator $D_B(U)$.

Bikerman: Ion concentrations

$$c_{-}^{(B)}(x) = \frac{c_b \exp(U(x))}{D_B(U(x))}$$
$$c_{+}^{(B)}(x) = \frac{c_b \exp(-U(x))}{D_B(U(x))}$$

Hence, the ion concentrations remain finite near highly charged surfaces.

SURFACE FREE CHARGE DENSITY: BIKERMAN

The surface free charge density is obtained from the electric field at the metal surface.

Surface free charge density

$$\sigma_{\text{free}} = -\varepsilon_0 \varepsilon_S \left. \frac{d\phi}{dx} \right|_{x=0^+}, \quad \sigma_{\text{free}} = 0 \iff \phi_M = \phi_{\text{pzc}}$$

CAPACITANCE: BIKERMAN

Let

$$\Delta\phi = \phi_M - \phi_{pzc}, \quad A = \frac{2\varepsilon_0\varepsilon_S k_B T}{r_0^3}, \quad B = 4r_0^3 N_A c_b$$

Bikerman differential capacitance

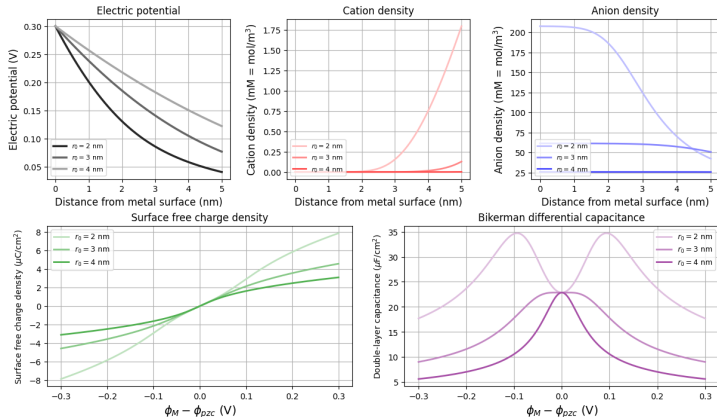
$$C_{dl}^{(B)}(\Delta\phi) = \begin{cases} \frac{\operatorname{sgn}(\Delta\phi) A B \frac{e_0}{2k_B T} \sinh\left(\frac{e_0 \Delta\phi}{k_B T}\right)}{2\sqrt{A \ln\left[1 + B \sinh^2\left(\frac{e_0 \Delta\phi}{2k_B T}\right)\right] \left[1 + B \sinh^2\left(\frac{e_0 \Delta\phi}{2k_B T}\right)\right]}}, & \Delta\phi \neq 0 \\ \varepsilon_0 \varepsilon_S \lambda_D^{-1}, & \Delta\phi = 0 \end{cases}$$

QUESTION 2: BIKERMAN

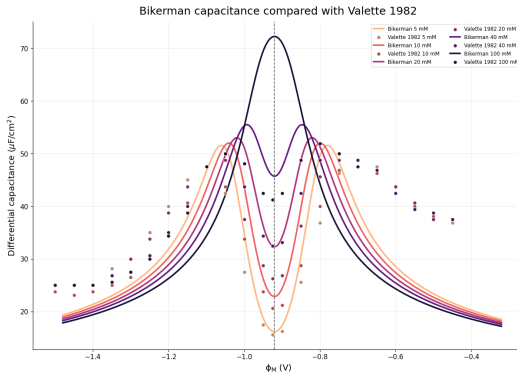
- 2.1) How does the electric potential change when varying the ion-size parameter r_0 ?
- 2.2) How does the Bikerman model differ from Gouy–Chapman near the metal surface?
- 2.3) How do the cation and anion concentrations change when r_0 increases?
- 2.4) Why do the concentrations no longer diverge at large applied potentials?
- 2.5) How does the finite-size correction affect the surface free charge density and the differential capacitance?
- 2.6) How well does the Bikerman capacitance reproduce the experimental Valette data?

The next two slides provide a numerical guide and a comparison with experiment.

QUESTION 2: NUMERICAL SOLUTION GUIDE



QUESTION 2: EXPERIMENTAL COMPARISON



Question: How well does the Bikerman capacitance reproduce the experimental Valette data? What is missing in the model that could improve the comparison?

FREISE: FUNDAMENTAL CONSTANTS

Symbol	Value	Unit	Meaning
N_A	6.022×10^{23}	mol^{-1}	Avogadro constant
k_B	8.617×10^{-5}	eV K^{-1}	Boltzmann constant
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FREISE: MODEL PARAMETERS

Symbol	Unit	Meaning
$\phi_M - \phi_{pzc}$	V	Applied potential relative to the Potential of Zero Charge.
T	K	Temperature
ϵ_S	–	Relative permittivity of the solution region
c_b	mM	Bulk ion concentration
r_0	nm	Reference ion / lattice size
γ_c	–	Cation volume factor
γ_a	–	Anion volume factor

MODEL SETUP: FREISE

The Freise model extends the Bikerman model by allowing cations and anions to have different effective volume coefficients.

In the symmetric Bikerman model, both ions have the same steric contribution. In the Freise model, this is generalized to

$$\gamma_c \neq \gamma_a.$$

This makes the positive and negative branches of the charge density and capacitance asymmetric with respect to the PZC.

MODEL SETUP: FREISE VARIABLES

Define the dimensionless potential and the volume parameter:

Dimensionless variables

$$U(x) = \frac{e_0 \phi(x)}{k_B T}, \quad \alpha = r_0^3 N_A c_b$$

$$D_F(U) = 1 + \alpha [\gamma_c (e^{-U} - 1) + \gamma_a (e^U - 1)]$$

The parameters γ_c and γ_a control the cation and anion volume contributions separately.

MODEL SETUP: FREISE EQUATION

The Freise model is described by a modified Poisson–Boltzmann equation with an asymmetric steric denominator.

Freise: Governing equation for ϕ

$$\varepsilon_0 \varepsilon_S \frac{d^2 \phi}{dx^2} = \frac{2e_0 N_A c_b \sinh\left(\frac{e_0 \phi}{k_B T}\right)}{1 + r_0^3 N_A c_b \left[\gamma_c \left(e^{-e_0 \phi / k_B T} - 1 \right) + \gamma_a \left(e^{+e_0 \phi / k_B T} - 1 \right) \right]}$$

with boundary conditions

$$\phi(0) = \phi_M - \phi_{pzc}, \quad \lim_{x \rightarrow \infty} \phi(x) = 0$$

MODEL SETUP: FREISE CONCENTRATIONS

The Freise concentrations are obtained from the Boltzmann factors divided by the asymmetric steric denominator $D_F(U)$.

Freise: Ion concentrations

$$c_+^{(F)}(x) = \frac{c_b e^{-U(x)}}{D_F(U(x))}$$
$$c_-^{(F)}(x) = \frac{c_b e^{+U(x)}}{D_F(U(x))}$$

For $\gamma_c = \gamma_a = 1$, the symmetric Bikerman model is recovered.

MODEL SETUP: DIFFERENT SIZE COEFFICIENTS

The two volume factors affect opposite branches of the double layer.

Physical interpretation

- γ_a controls the anion volume contribution. It is most visible near a positively charged metal surface.
- γ_c controls the cation volume contribution. It is most visible near a negatively charged metal surface.
- If $\gamma_c \neq \gamma_a$, the model is no longer symmetric around $\phi_M - \phi_{pzc} = 0$.

SURFACE CHARGE DENSITY: FREISE

The surface free charge density is obtained from the electric field at the metal surface.

Surface free charge density

$$\sigma_{\text{free}} = -\epsilon_0 \epsilon_S \left. \frac{d\phi}{dx} \right|_{x=0^+}, \quad \sigma_{\text{free}} = 0 \iff \phi_M = \phi_{\text{pzc}}.$$

CAPACITANCE: FREISE

The differential capacitance is obtained by differentiating the Freise surface charge density with respect to the applied potential.

Freise differential capacitance

$$C_{\text{dl}}^{(\text{F})}(\phi_{\text{M}} - \phi_{\text{pzc}}) = \frac{d\sigma_{\text{free}}^{(\text{F})}}{d(\phi_{\text{M}} - \phi_{\text{pzc}})}.$$

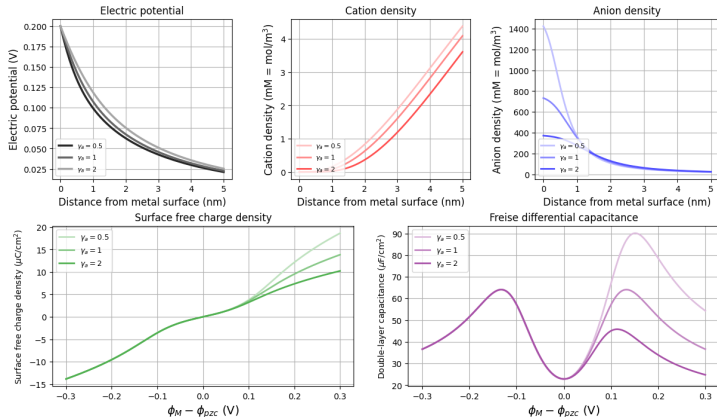
If $\gamma_{\text{c}} \neq \gamma_{\text{a}}$, the capacitance is asymmetric around the PZC.

QUESTION 3: DIFFERENT SIZE COEFFICIENTS (EXTRA)

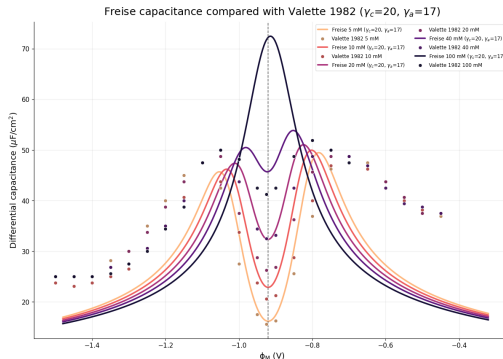
Optional / extra: Freise extends Bikerman by allowing different cation and anion volume factors.

- 3.1) What changes when $\gamma_c \neq \gamma_a$?
- 3.2) How does increasing γ_a affect anions near a positively charged surface?
- 3.3) How does increasing γ_c affect cations near a negatively charged surface?
- 3.4) Why does σ_{free} become asymmetric with respect to $\phi_M - \phi_{\text{pzc}} = 0$?
- 3.5) Why does C_{dl} become asymmetric when $\gamma_c \neq \gamma_a$?
- 3.6) Does Freise improve agreement with Valette data compared with symmetric Bikerman?

QUESTION 3: NUMERICAL SOLUTION GUIDE (EXTRA)



QUESTION 3: EXPERIMENTAL COMPARISON (EXTRA)



Question: How well does the asymmetric Freise capacitance reproduce the experimental Valette data? Does allowing different cation and anion volume factors improve the comparison compared with the symmetric Bikerman model?